

Hint: The 3-dB frequency of a cascade of N identical gain stages, each with a 3-dB frequency $f_{3\text{dB}}|_{\text{stage}}$ is given by

$$f_{3\text{dB}}|_{\text{cascade}} = f_{3\text{dB}}|_{\text{stage}} \sqrt{2^{1/N} - 1}$$

D 11.22 Design a supply-ripple-reduced power amplifier for which the output stage can be modeled by the block diagram of Fig. 11.5, where $A_1 = 0.9 \text{ V/V}$, and the power-supply ripple $V_N = \pm 1 \text{ V}$. A closed-loop gain of 10 V/V is desired. What is the gain of the low-ripple preamplifier needed to reduce the output ripple to $\pm 100 \text{ mV}$? To $\pm 10 \text{ mV}$? To $\pm 1 \text{ mV}$? For each case, specify the value required for the feedback factor β .

D 11.23 A feedback amplifier is to be designed using a feedback loop connected around a two-stage amplifier. The first stage is a direct-coupled, small-signal amplifier with a high upper 3-dB frequency. The second stage is a power-output stage with a midband gain of 10 V/V and upper and lower 3-dB frequencies of 8 kHz and 80 Hz , respectively. The feedback amplifier should have a midband gain of 100 V/V and an upper 3-dB frequency of 40 kHz . What is the required gain of the small-signal amplifier? What value of β should be used? What does the lower 3-dB frequency of the overall amplifier become?

***11.24** The complementary BJT follower shown in Fig. P11.24(a) has the approximate transfer characteristic shown in Fig. P11.24(b). Observe that for $-0.7 \text{ V} \leq v_i \leq +0.7 \text{ V}$, the output is zero. This “dead band” leads to crossover distortion (see Section 12.3). Consider this follower to be driven by the output of a differential amplifier of gain 100

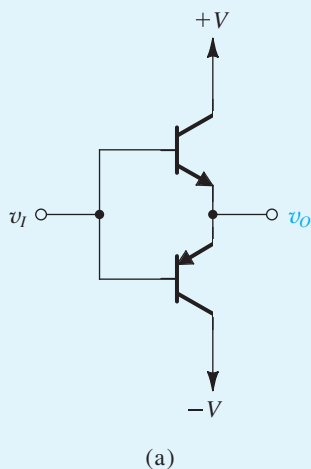


Figure P11.24

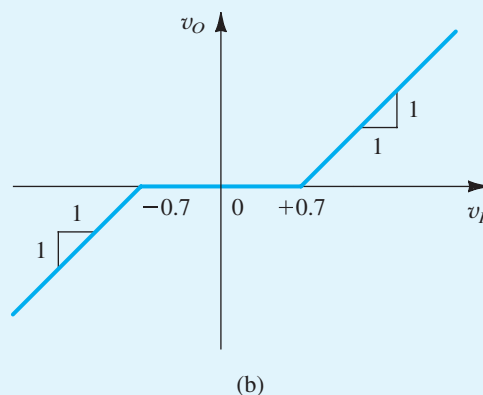


Figure P11.24 continued

whose positive-input terminal is connected to the input signal source v_s and whose negative-input terminal is connected to the emitters of the follower. Sketch the transfer characteristic v_o versus v_s of the resulting feedback amplifier. What are the limits of the dead band, and what are the gains outside the dead band?

D 11.25 A particular amplifier has a nonlinear transfer characteristic that can be approximated as follows:

- For small input signals, $|v_i| \leq 10 \text{ mV}$, $v_o/v_i = 10^3$.
- For intermediate input signals, $10 \text{ mV} \leq |v_i| \leq 60 \text{ mV}$, $\Delta v_o/\Delta v_i = 10^2$.
- For large input signals, $|v_i| \geq 60 \text{ mV}$, the output saturates.

If the amplifier is connected in a negative-feedback loop, find the feedback factor β that reduces the factor-of-10 change in gain (occurring at $|v_i| = 10 \text{ mV}$) to only a 10% change. What is the transfer characteristic v_o versus v_s of the amplifier with feedback?

Section 11.3: The Feedback Voltage Amplifier

D 11.26 For the feedback voltage amplifier of Fig. 11.8(a), let the op amp have an infinite input resistance, a zero output resistance, and a finite open-loop gain of 1000 V/V . If $R_1 = 10 \text{ k}\Omega$, what value should R_2 have to obtain an ideal closed-loop gain of 10? Now, calculate the loop gain $A\beta$ and use it to find the actual value of the closed-loop gain A_f . If A_f is to be exactly 10, what must the value of R_2 be?

D 11.27 Consider the series–shunt feedback amplifier in Fig. 11.11(a), which is analyzed in Example 11.3.

- If $R_1 = 10 \text{ k}\Omega$, find the value of R_2 that results in an ideal closed-loop gain of 10.

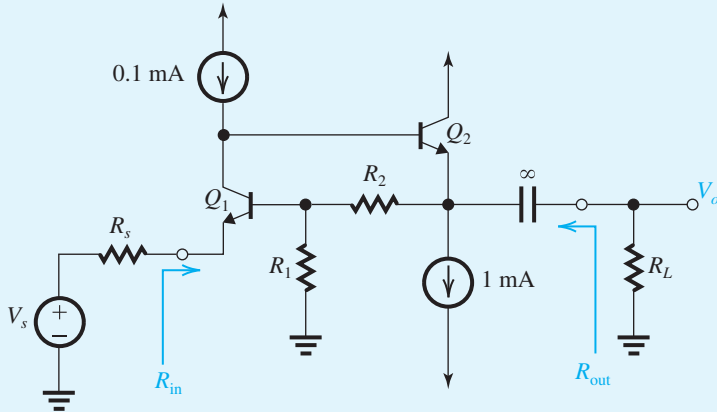


Figure P11.29

- (b) Use the expression for $A\beta$ derived in Example 11.3 to find the value of the loop gain for the case $\mu = 1000$, $R_{id} = 100 \text{ k}\Omega$, $r_o = 1 \text{ k}\Omega$, $R_s = 100 \text{ k}\Omega$, and $R_L = 10 \text{ k}\Omega$. Hence determine the value of the closed-loop gain A_f .
- (c) By what factor must μ be increased to ensure that A_f is within 1% of the ideal value of 10?

D 11.28 Consider the series–shunt feedback amplifier of Fig. 11.8(b) that is analyzed in Example 11.2.

- (a) If $R_1 = 1 \text{ k}\Omega$, what value should R_2 have to obtain a closed-loop gain whose ideal value is 5 V/V?
- (b) If $g_{m1} = g_{m2} = 4 \text{ mA/V}$, $R_{D1} = R_{D2} = 10 \text{ k}\Omega$, and the MOSFET's r_o is very large, use the expression for $A\beta$ derived in Example 11.2 to find the value of $A\beta$ and hence determine the closed-loop gain A_f .

***11.29** In the series–shunt feedback amplifier shown in Fig. P11.29, the devices operate with $V_{BE} = 0.7 \text{ V}$ and have $\beta_1 = \beta_2 = 100$. The input signal V_s has a zero dc component. Resistances $R_s = 100 \text{ }\Omega$, $R_1 = 1 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, and $R_L = 1 \text{ k}\Omega$.

- (a) If the loop gain is large, what do you expect the closed-loop gain to be? Give both an expression and its value.
- (b) Find the dc emitter current in each of Q_1 and Q_2 . Also, find the dc voltage at the emitter of Q_2 .
- (c) Calculate the value of the loop gain $A\beta$. (Hint: Set $V_s = 0$ and break the loop at the base of Q_1 . Simplify the circuit by eliminating dc sources.)
- (d) Calculate the value of A_f .

D 11.30 Consider the series–shunt feedback amplifier of Fig. 11.8(c), which was the subject of Exercise 11.6. Assume that the voltage divider (R_1, R_2) is implemented with a 1-M Ω potentiometer. Assume that the MOSFET is biased so that $g_m = 4 \text{ mA/V}$ and r_o is large. Also, $R_D = 10 \text{ k}\Omega$. Find the value of R_1 that results in a closed-loop gain of 5 V/V.

D 11.31 Figure P11.31 shows a series–shunt feedback amplifier known as a “feedback triple.” All three MOSFETs are biased to operate at $g_m = 4 \text{ mA/V}$. You may neglect their r_o 's.

- (a) Select a value for R_F that results in a closed-loop gain that is ideally 10 V/V.

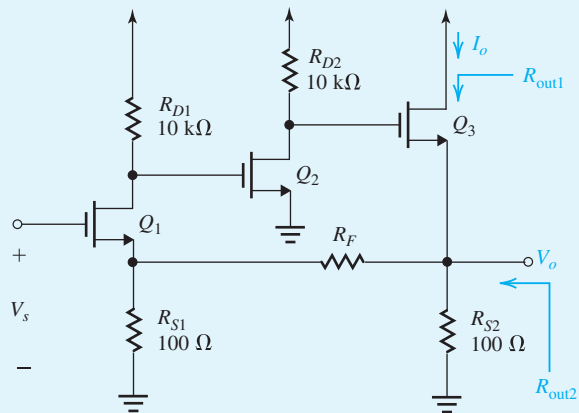


Figure P11.31

- (b) Determine the loop gain $A\beta$ and hence the value of A_f . By what percentage does A_f differ from the ideal value you designed for? How can you adjust the circuit to make A_f equal to 10?

11.32 Figure P11.32 shows a series–shunt feedback amplifier without details of the bias circuit.

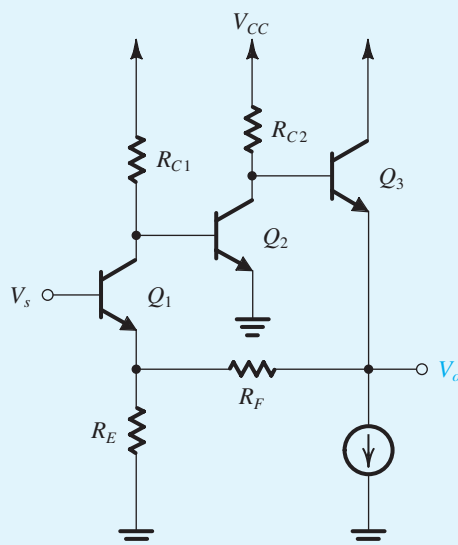


Figure P11.32

- (a) If R_E is selected to be $50\ \Omega$, find the value for R_F that results in a closed-loop gain with an ideal value of $25\ \text{V/V}$.
- (b) If Q_1 is biased at $1\ \text{mA}$, Q_2 at $2\ \text{mA}$, and Q_3 at $5\ \text{mA}$, and assuming that the transistors have $h_{fe} = 100$ and large r_o , and that $R_{C1} = 2\ \text{k}\Omega$ and $R_{C2} = 1\ \text{k}\Omega$, find the value of the loop gain $A\beta$ and hence of the closed-loop gain A_f .

D 11.33 The current-mirror-loaded differential amplifier in Fig. P11.33 has a feedback network consisting of the voltage divider (R_1, R_2), with $R_1 + R_2 = 1\ \text{M}\Omega$. The devices are sized to operate at $|V_{OV}| = 0.2\ \text{V}$. For all devices, $|V_A| = 10\ \text{V}$. The input signal source has a zero dc component.

- (a) Find the loop gain $A\beta$ and hence the value of A .
- (b) Find the values of R_1 and R_2 that result in a closed-loop gain of exactly $5\ \text{V/V}$.

Section 11.4: Systematic Analysis of Feedback Voltage Amplifiers (Series–Shunt)

11.34 A series–shunt feedback amplifier employs a basic amplifier with input and output resistances each of $2\ \text{k}\Omega$ and gain $A = 1000\ \text{V/V}$. The feedback factor $\beta = 0.1\ \text{V/V}$. Find

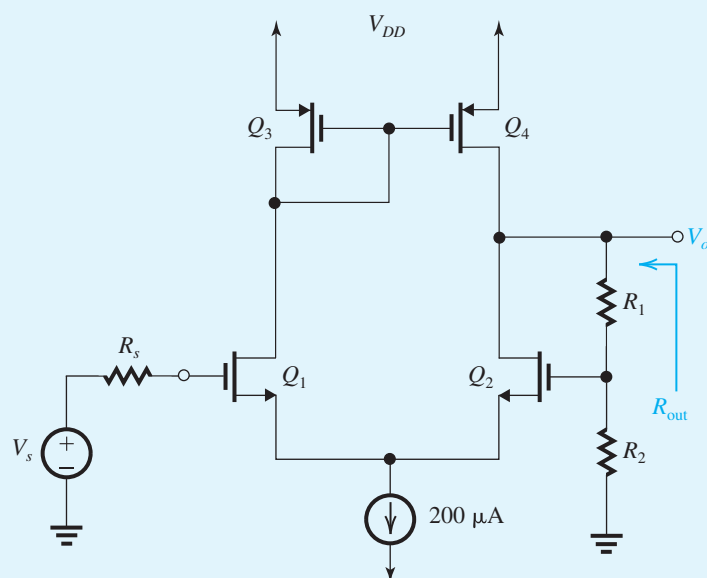


Figure P11.33

the gain A_f , the input resistance R_{if} , and the output resistance R_{of} of the closed-loop amplifier.

11.35 For a particular amplifier connected in a feedback loop in which the output voltage is sampled, measurement of the output resistance before and after the loop is connected shows a change by a factor of 200. Is the resistance with feedback higher or lower? What is the value of the loop gain $A\beta$? If R_{of} is $100\ \Omega$, what is R_o without feedback?

11.36 The formulas for R_{if} and R_{of} in Eqs. (11.20) and (11.23), respectively, also apply for the case in which A is a function of frequency. In this case, the resulting impedances Z_{if} and Z_{of} will be functions of frequency. Consider the case of a series–shunt amplifier that has an input resistance R_i , an output resistance R_o , and open-loop gain $A = A_0/[1 + (s/\omega_H)]$, and a feedback factor β that is independent of frequency. Find Z_{if} and Z_{of} and give an equivalent circuit for each, together with the values of all the elements in the equivalent circuits.

11.37 A feedback amplifier utilizing voltage sampling and employing a basic voltage amplifier with a gain of $1000\ \text{V/V}$ and an input resistance of $1000\ \Omega$ has a closed-loop input resistance of $10\ \text{k}\Omega$. What is the closed-loop gain? If the basic amplifier is used to implement a unity-gain voltage buffer, what input resistance do you expect?

11.38 Consider the noninverting op-amp circuit of Example 11.4 for the case $R_1 = \infty$ and $R_2 = 0$.

- What is the value of β , and what is the ideal value of the closed-loop gain?
- Adapt the expressions found in Example 11.4 to obtain expressions for A and $A\beta$ for this case.
- For $\mu = 10^4$, $R_{id} = 100\ \text{k}\Omega$, $R_s = 10\ \text{k}\Omega$, $r_o = 1\ \text{k}\Omega$, and $R_L = 2\ \text{k}\Omega$, find A , $A\beta$, A_f , R_{in} , and R_{out} .

***11.39** This problem deals with the series–shunt feedback amplifier of Fig. P11.29 and overlaps somewhat with Problem 11.29. Thus, if you have already solved 11.29, you can use some of the results in the solution of this problem. The devices operate with $V_{BE} = 0.7\ \text{V}$ and have $\beta_1 = \beta_2 = 100$. The input signal V_s has a zero dc component. Resistances $R_s = 100\ \Omega$, $R_1 = 1\ \text{k}\Omega$, $R_2 = 10\ \text{k}\Omega$, and $R_L = 1\ \text{k}\Omega$.

- If the loop gain is large, what do you expect the closed-loop gain V_o/V_s to be? Give both an expression and its approximate value.

- Find the dc emitter current in each of Q_1 and Q_2 . Also find the dc voltage at the emitter of Q_2 .
- Sketch the A circuit without the dc sources. Derive expressions for A , R_i , and R_o , and find their values.
- Give an expression for β and find its value.
- Find the closed-loop gain V_o/V_s , the input resistance R_{in} , and the output resistance R_{out} . By what percentage does the value of A_f differ from the approximate value found in (a)?

SIM D *11.40 Figure P11.40 shows a series–shunt amplifier with a feedback factor $\beta = 1$. The amplifier is designed so that $v_o = 0$ for $v_s = 0$, with small deviations in v_o from $0\ \text{V}$ dc being minimized by the negative-feedback action. The technology utilized has $k'_n = 2k'_p = 120\ \mu\text{A/V}^2$, $|V_t| = 0.7\ \text{V}$, and $|V_A| = 24\ \text{V}/\mu\text{m}$.

- Show that the feedback is negative.
- With the feedback loop opened at the gate of Q_2 , and the gate terminals of Q_1 and Q_2 grounded, find the dc current and the overdrive voltage at which each of Q_1 to Q_5 is operating. Ignore the Early effect. Also find the dc voltage at the output.
- Find g_m and r_o of each of the five transistors.
- Find expressions and values of A and R_o . Assume that the bias current sources are ideal.
- Find the gain with feedback, A_f , and the output resistance R_{out} .
- How would you modify the circuit to realize a closed-loop voltage gain of $5\ \text{V/V}$? What is the value of output resistance obtained?

***11.41** Figure P11.41 shows a series–shunt amplifier in which the three MOSFETs are sized to operate at $|V_{ov}| = 0.2\ \text{V}$. Let $|V_t| = 0.5\ \text{V}$ and $|V_A| = 10\ \text{V}$. The current sources utilize single transistors and thus have output resistances equal to r_o .

- Show that the feedback is negative.
- Assuming the loop gain to be large, what do you expect the closed-loop voltage gain V_o/V_s to be approximately?
- If V_s has a zero dc component, find the dc voltages at nodes S_1 , G_2 , S_3 , and G_3 . Verify that each of the current sources has the minimum required dc voltage across it for proper operation.
- Find the A circuit. Calculate the gain of each of the three stages and the overall voltage gain, A .

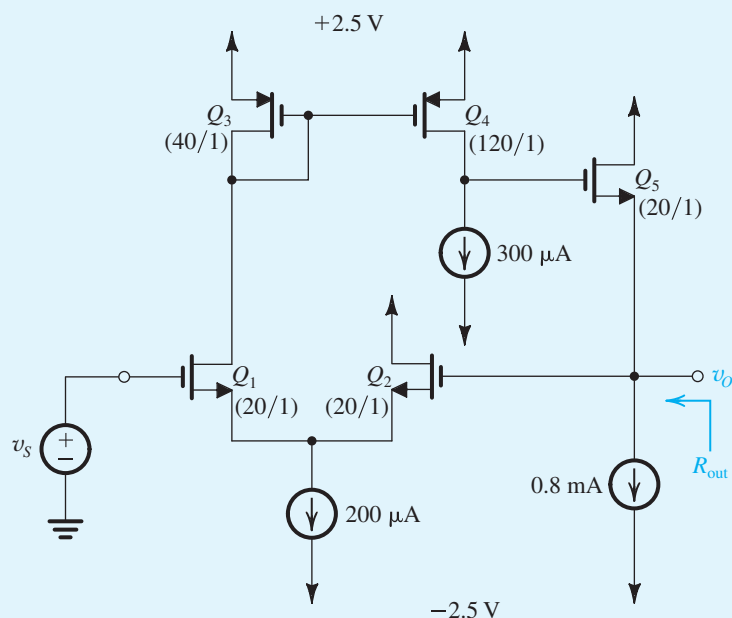


Figure P11.40

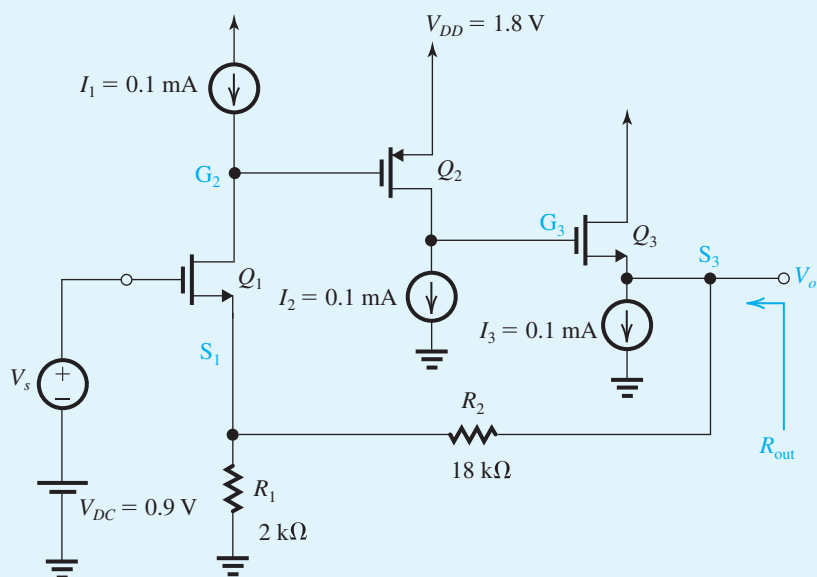


Figure P11.41

- [Hint: A CS amplifier with a resistance R_s in the source lead has an effective transconductance $g_m/(1 + g_m R_s)$ and an output resistance $r_o(1 + g_m R_s)$.]
- (e) Find β .
- (f) Find $A_f = V_o/V_s$. By what percentage does this value differ from the approximate value obtained in (b)?
- (g) Find the output resistance R_{out} .

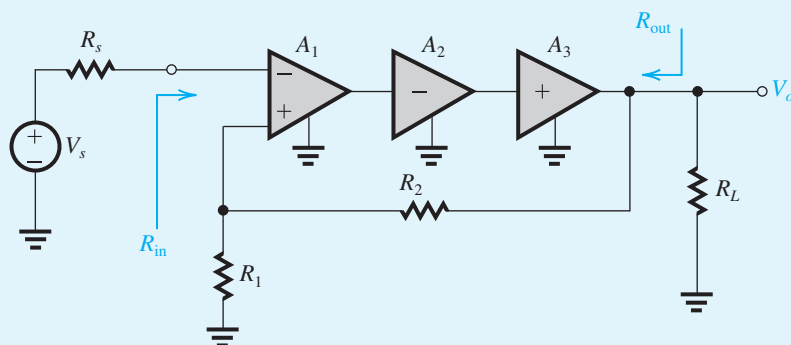


Figure P11.45

feedback whose amount is controlled by the setting of the wiper. A voltage signal V_s is applied between the noninverting input (+In) and ground. A load resistance $R_L = 100 \text{ k}\Omega$ is connected between the output terminal and ground. The potentiometer is adjusted to obtain a closed-loop gain $A_f \equiv V_o/V_s \approx 10 \text{ V/V}$.

Specify the required setting of the potentiometer by giving the values of R_1 and R_2 . Toward this end, find the A circuit (supply a circuit diagram), the value of A , the β circuit (supply a circuit diagram), and the value of β .

- (f) What is the output resistance of the feedback amplifier, excluding R_L ?

D *11.44 Figure P11.32 shows a series–shunt feedback amplifier without details of the bias circuit.

- (a) Eliminating the dc sources, sketch the A circuit and the circuit for determining β .
 (b) Show that if $A\beta$ is large then the closed-loop voltage gain is given approximately by

$$A_f \equiv \frac{V_o}{V_s} \simeq \frac{R_F + R_E}{R_E}$$

- (c) If R_E is selected equal to 50Ω , find R_F that will result in a closed-loop gain of approximately 25 V/V .
 (d) If Q_1 is biased at 1 mA , Q_2 at 2 mA , and Q_3 at 5 mA , and assuming that the transistors have $h_{fe} = 100$, find approximate values for R_{C1} and R_{C2} to obtain gains from the stages of the A circuit as follows: a voltage gain of Q_1 of about -10 and a voltage gain of Q_2 of about -50 .

- (e) For your design, what is the closed-loop voltage gain realized?
 (f) Calculate the input and output resistances of the closed-loop amplifier designed.

D *11.45 Figure P11.45 shows a three-stage feedback amplifier:

A_1 has an $82\text{-k}\Omega$ differential input resistance, a 20-V/V open-circuit differential voltage gain, and a $3.2\text{-k}\Omega$ output resistance.

A_2 has a $5\text{-k}\Omega$ input resistance, a 20-mA/V short-circuit transconductance, and a $20\text{-k}\Omega$ output resistance.

A_3 has a $20\text{-k}\Omega$ input resistance, unity open-circuit voltage gain, and a $1\text{-k}\Omega$ output resistance.

The feedback amplifier feeds a $1\text{-k}\Omega$ load resistance and is fed by a signal source with a $9\text{-k}\Omega$ resistance.

- (a) Show that the feedback is negative.
 (b) If $R_1 = 20 \text{ k}\Omega$, find the value of R_2 that results in a closed-loop gain V_o/V_s that is ideally 5 V/V .
 (c) Supply the small-signal equivalent circuit.
 (d) Sketch the A circuit and determine A .
 (e) Find β and the amount of feedback.
 (f) Find the closed-loop gain $A_f \equiv V_o/V_s$.
 (g) Find the feedback amplifier's input resistance R_{in} .
 (h) Find the feedback amplifier's output resistance R_{out} .
 (i) If the high-frequency response of the open-loop gain A is dominated by a pole at 100 Hz , what is the upper 3-dB frequency of the closed-loop gain?
 (j) If for some reason A_1 drops to half its nominal value, what is the percentage change in A_f ?